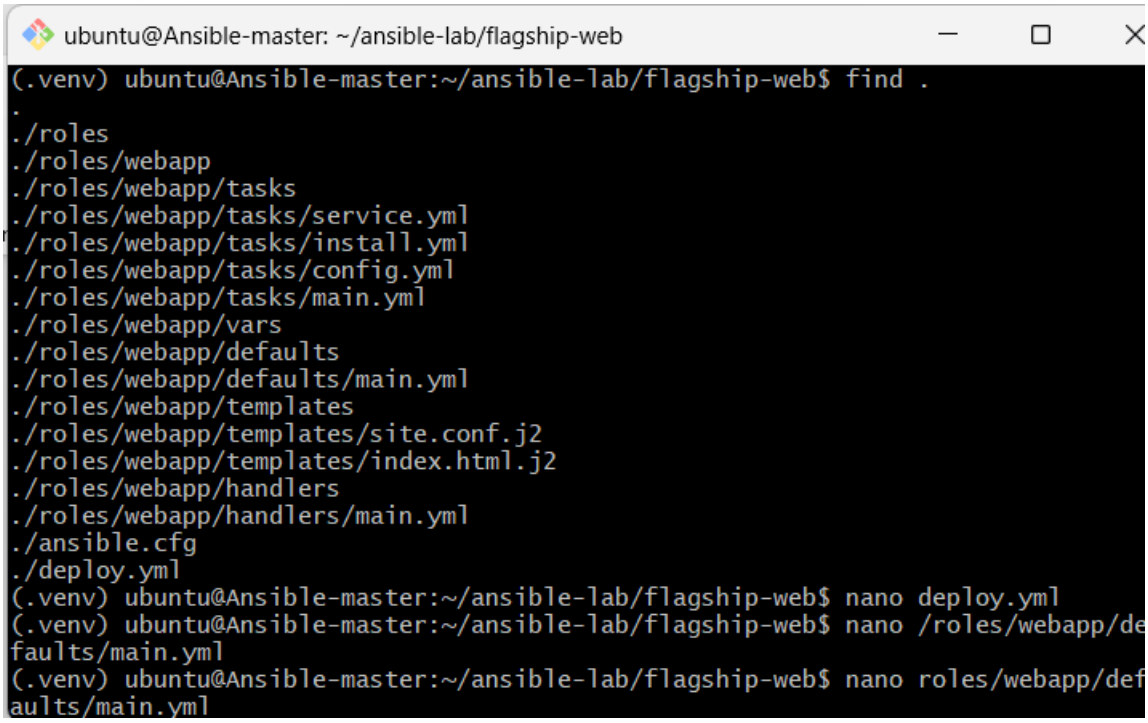
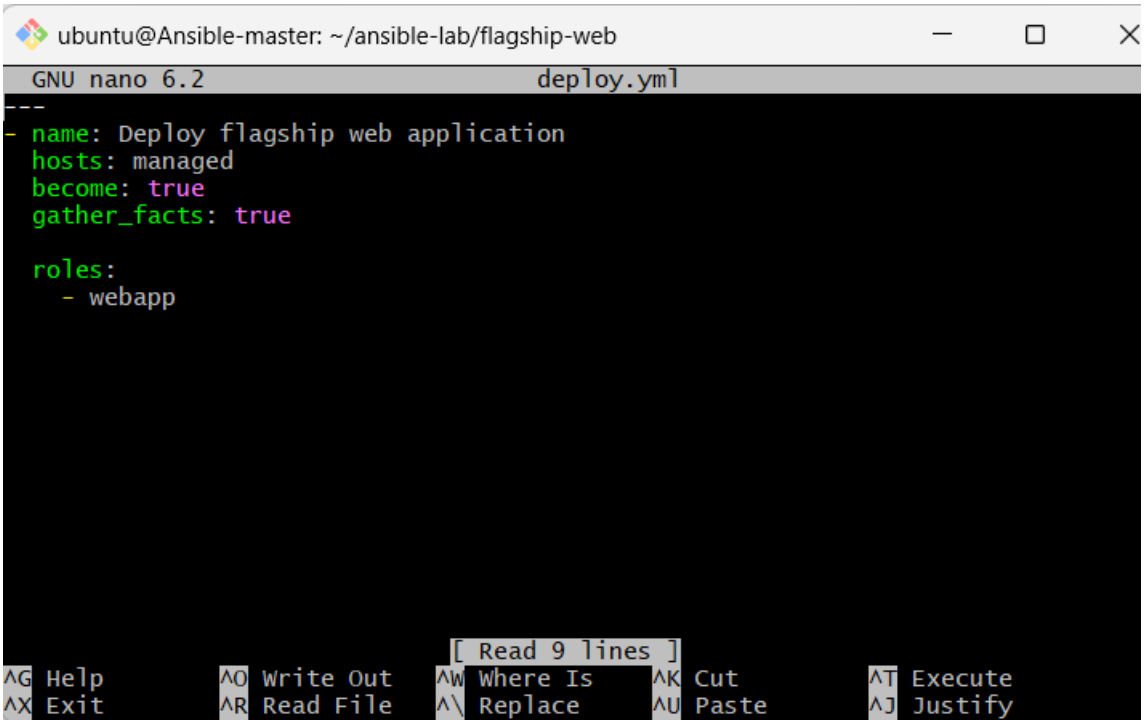
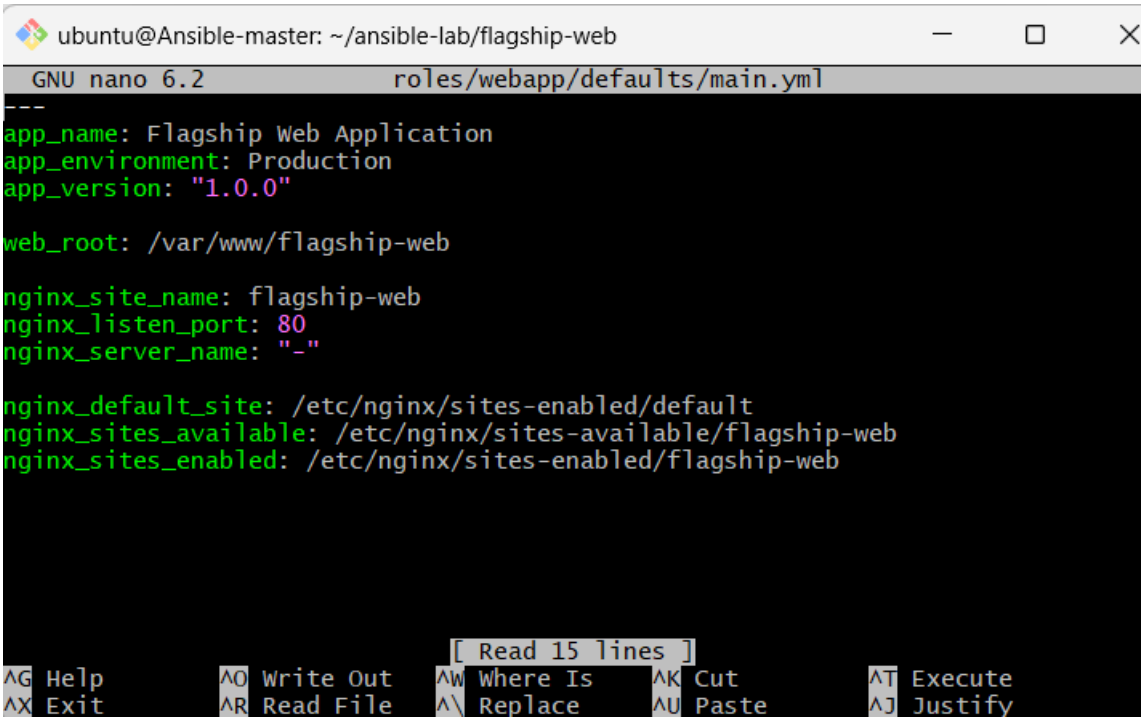
Ansible Mastery Part-5(Flagship Practical-1)

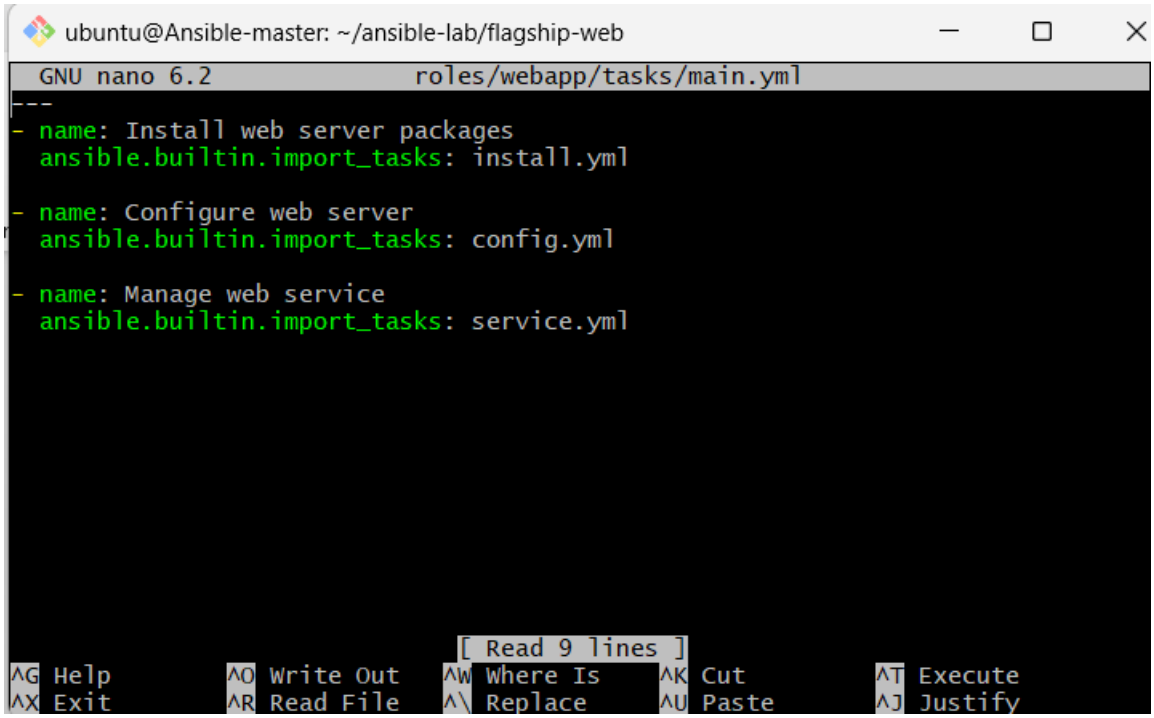
Enterprise-Grade Role-Based Web Application Deployment and Configuration Management with Ansible on AWS EC2 : Modular Roles, Jinja2 Templates, Idempotent Tasks, Event-Driven Handlers, Nginx Configuration Validation, Automated Service Reloads, and End-to-End HTTP Health Checks



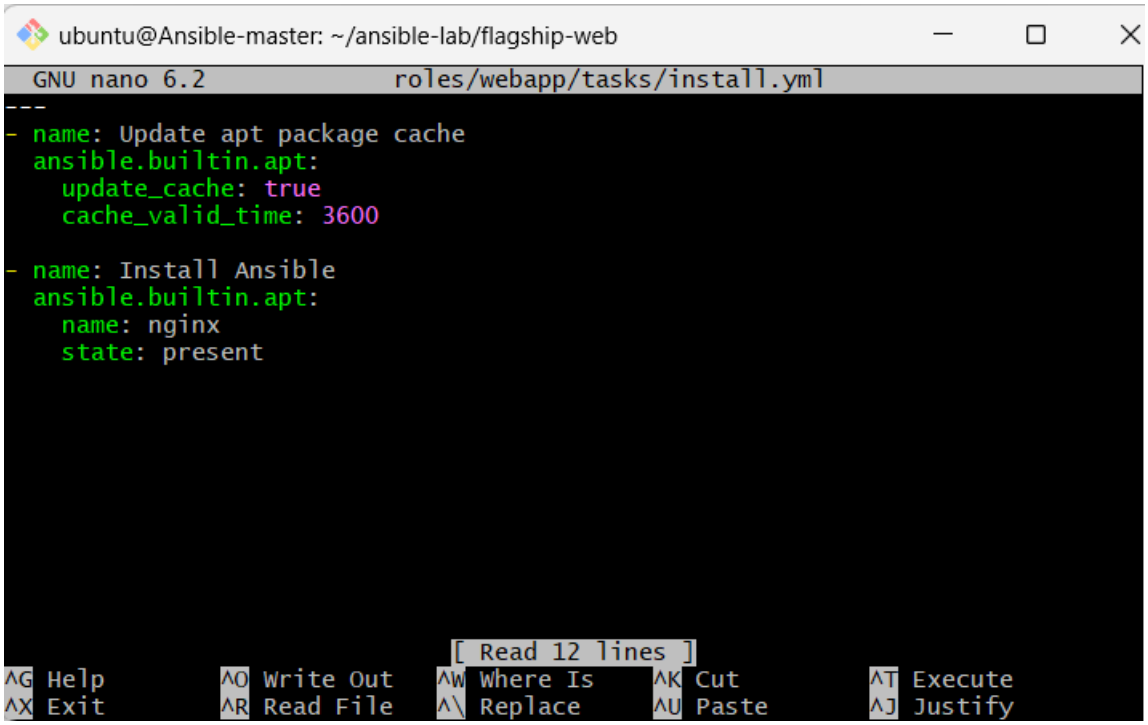

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
GNU nano 6.2 deploy.yml

- name: Deploy flagship web application
hosts: managed
become: true
gather_facts: true
roles:
- webapp
[Read 9 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

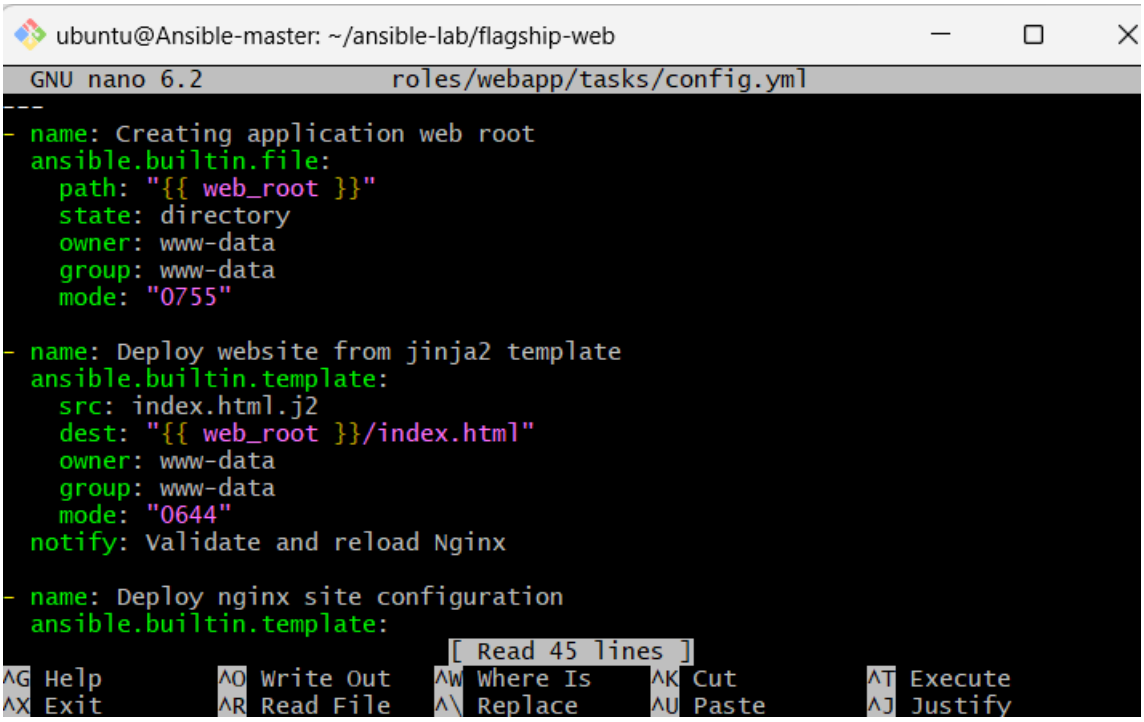

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
GNU nano 6.2 roles/webapp/defaults/main.yml
---
app_name: Flagship Web Application
app_environment: Production
app_version: "1.0.0"
web_root: /var/www/flagship-web
nginx_site_name: flagship-web
nginx_listen_port: 80
nginx_server_name: "_"
nginx_default_site: /etc/nginx/sites-enabled/default
nginx_sites_available: /etc/nginx/sites-available/flagship-web
nginx_sites_enabled: /etc/nginx/sites-enabled/flagship-web
[Read 15 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify


ubuntu@Ansible-master: ~/ansible-lab/flagship-web
GNU nano 6.2 roles/webapp/tasks/main.yml

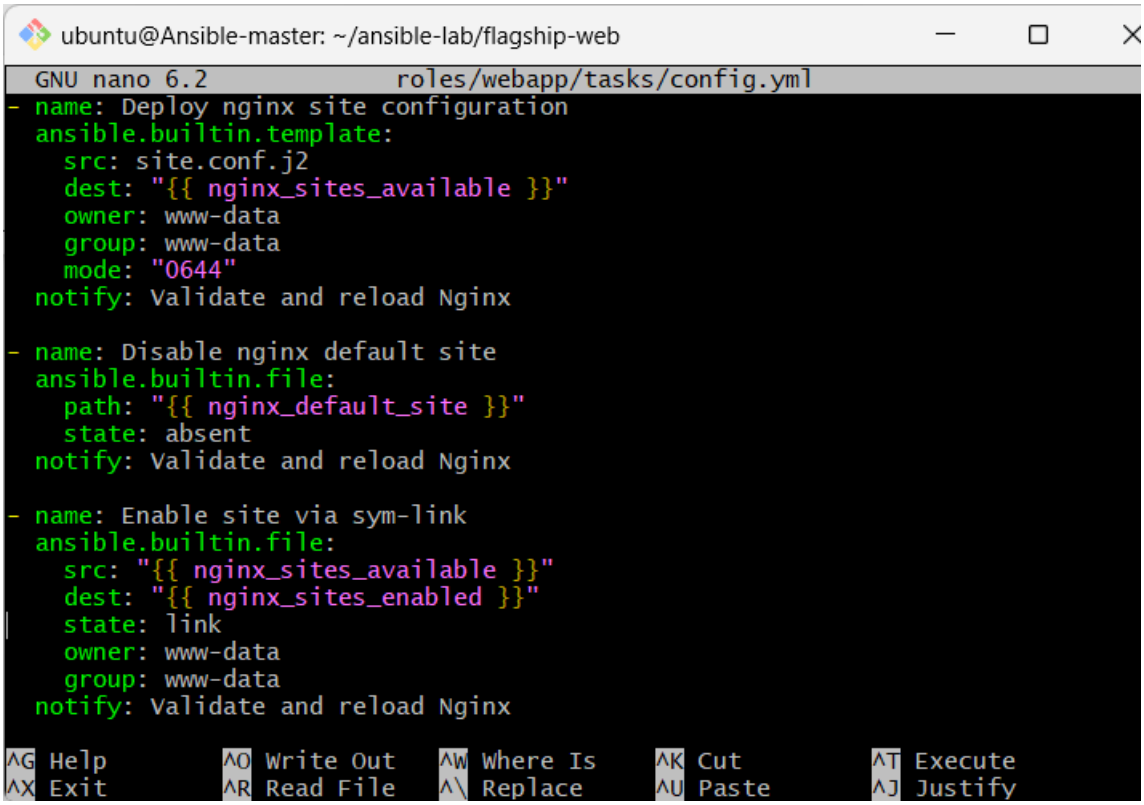
- name: Install web server packages
ansible.builtin.import_tasks: install.yml
- name: Configure web server
ansible.builtin.import_tasks: config.yml
- name: Manage web service
ansible.builtin.import_tasks: service.yml
[Read 9 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

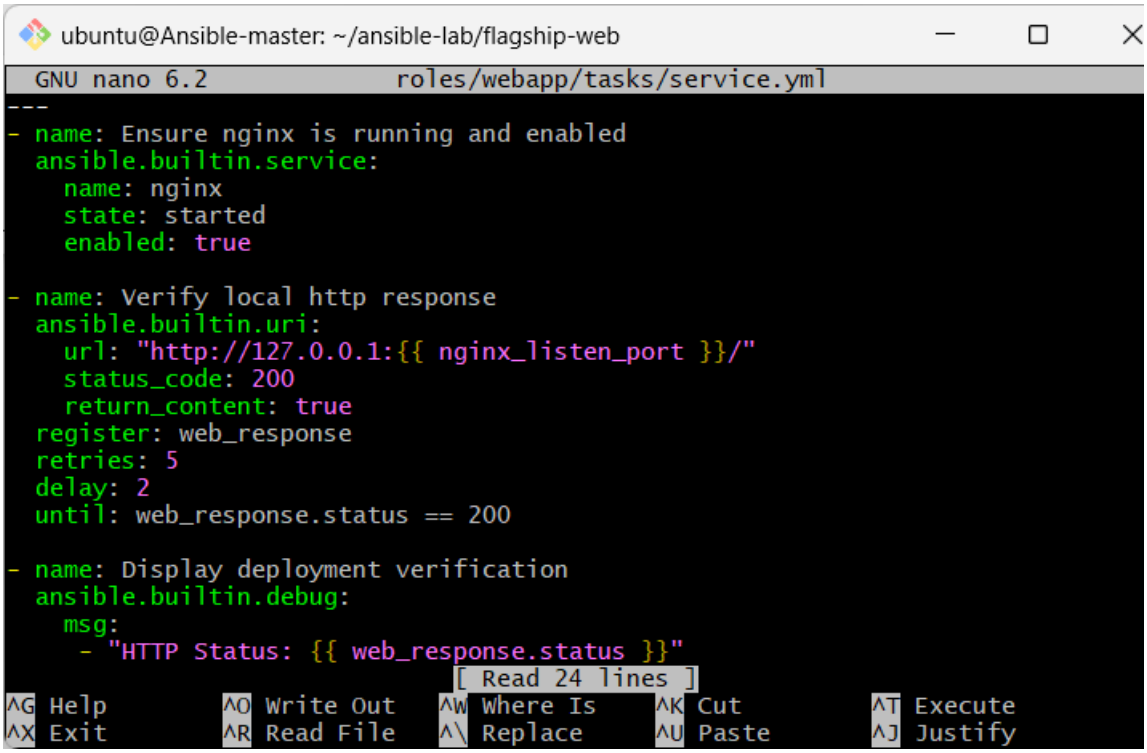

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
GNU nano 6.2 roles/webapp/tasks/install.yml

- name: Update apt package cache
ansible.builtin.apt:
update_cache: true
cache_valid_time: 3600
- name: Install Ansible
ansible.builtin.apt:
name: nginx
state: present
[Read 12 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

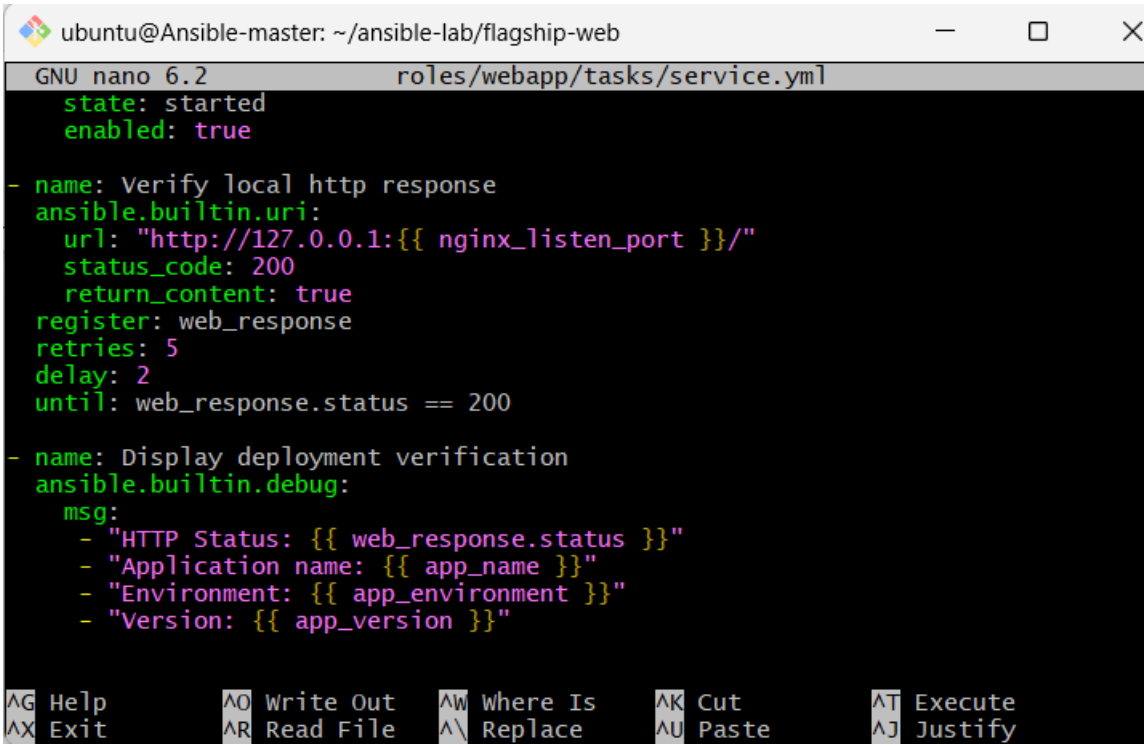

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
GNU nano 6.2 roles/webapp/tasks/config.yml

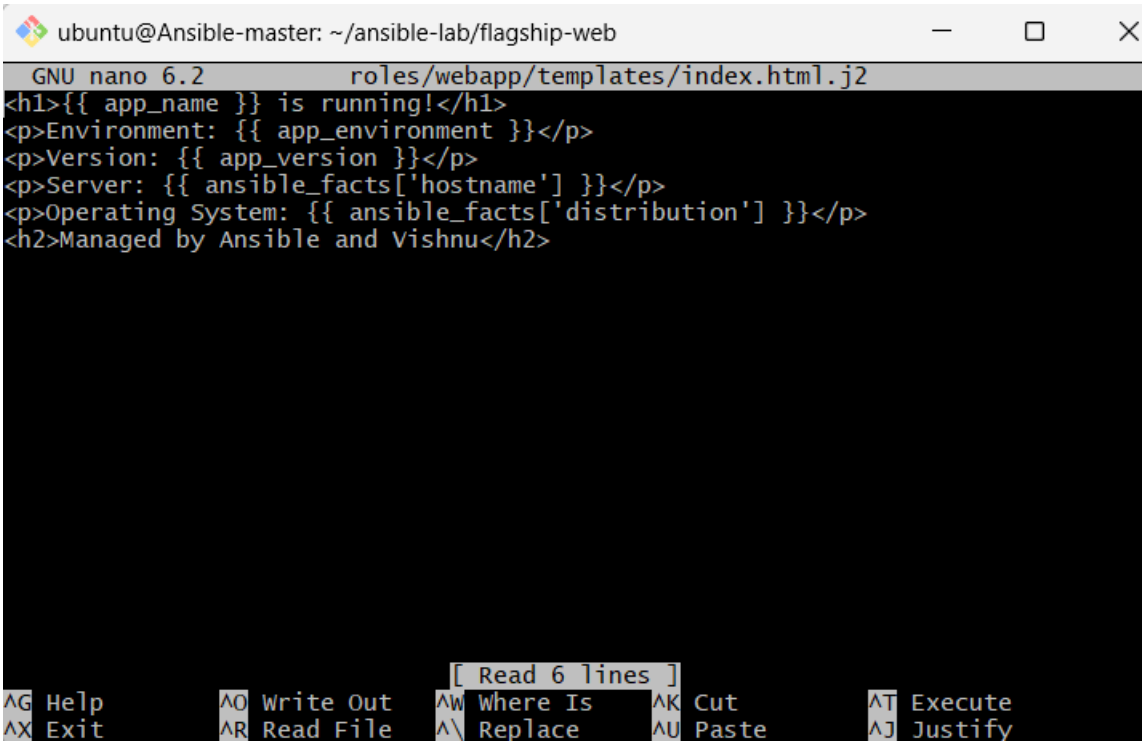
- name: Creating application web root
ansible.builtin.file:
path: "{{ web_root }}"
state: directory
owner: www-data
group: www-data
mode: "0755"
- name: Deploy website from jinja2 template
ansible.builtin.template:
src: index.html.j2
dest: "{{ web_root }}/index.html"
owner: www-data
group: www-data
mode: "0644"
notify: Validate and reload Nginx
- name: Deploy nginx site configuration
ansible.builtin.template:
[Read 45 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

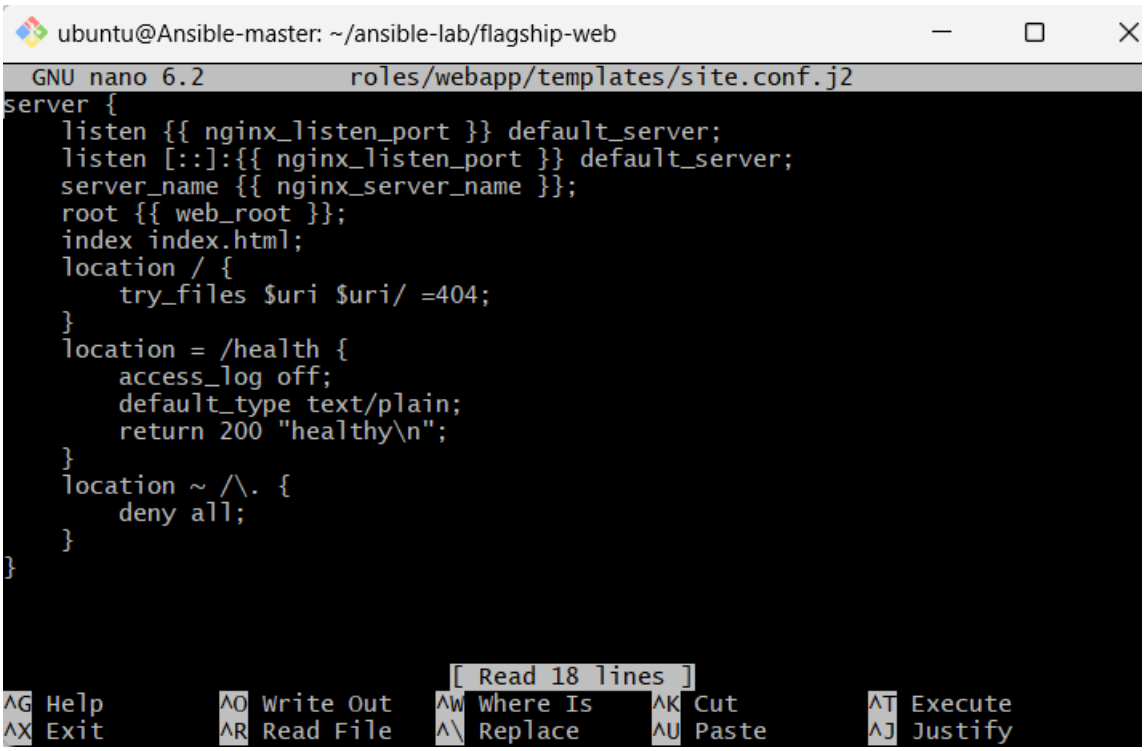

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
GNU nano 6.2 roles/webapp/tasks/config.yml
- name: Deploy nginx site configuration
ansible.builtin.template:
src: site.conf.j2
dest: "{{ nginx_sites_available }}"
owner: www-data
group: www-data
mode: "0644"
notify: Validate and reload Nginx
- name: Disable nginx default site
ansible.builtin.file:
path: "{{ nginx_default_site }}"
state: absent
notify: Validate and reload Nginx
- name: Enable site via sym-link
ansible.builtin.file:
src: "{{ nginx_sites_available }}"
dest: "{{ nginx_sites_enabled }}"
state: link
owner: www-data
group: www-data
notify: Validate and reload Nginx
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

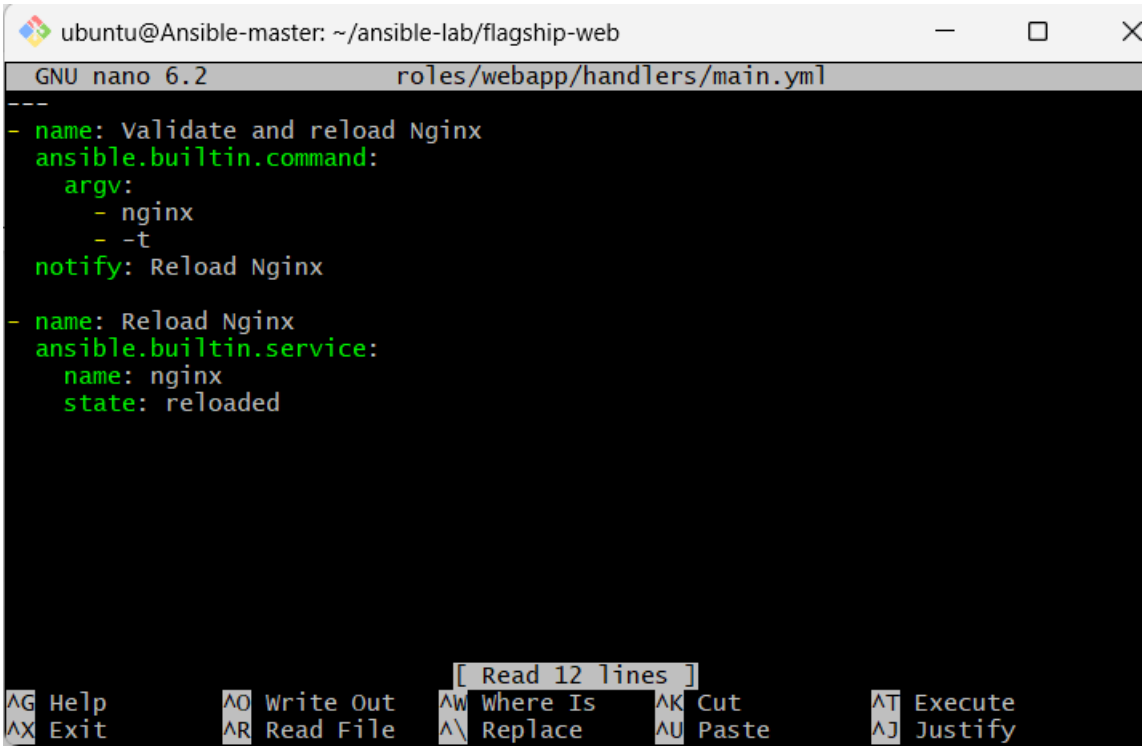

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
GNU nano 6.2 roles/webapp/tasks/service.yml

- name: Ensure nginx is running and enabled
ansible.builtin.service:
name: nginx
state: started
enabled: true
- name: Verify local http response
ansible.builtin.uri:
url: "http://127.0.0.1:{{ nginx_listen_port }}/"
status_code: 200
return_content: true
register: web_response
retries: 5
delay: 2
until: web_response.status == 200
- name: Display deployment verification
ansible.builtin.debug:
msg:
- "HTTP Status: {{ web_response.status }}"
[Read 24 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

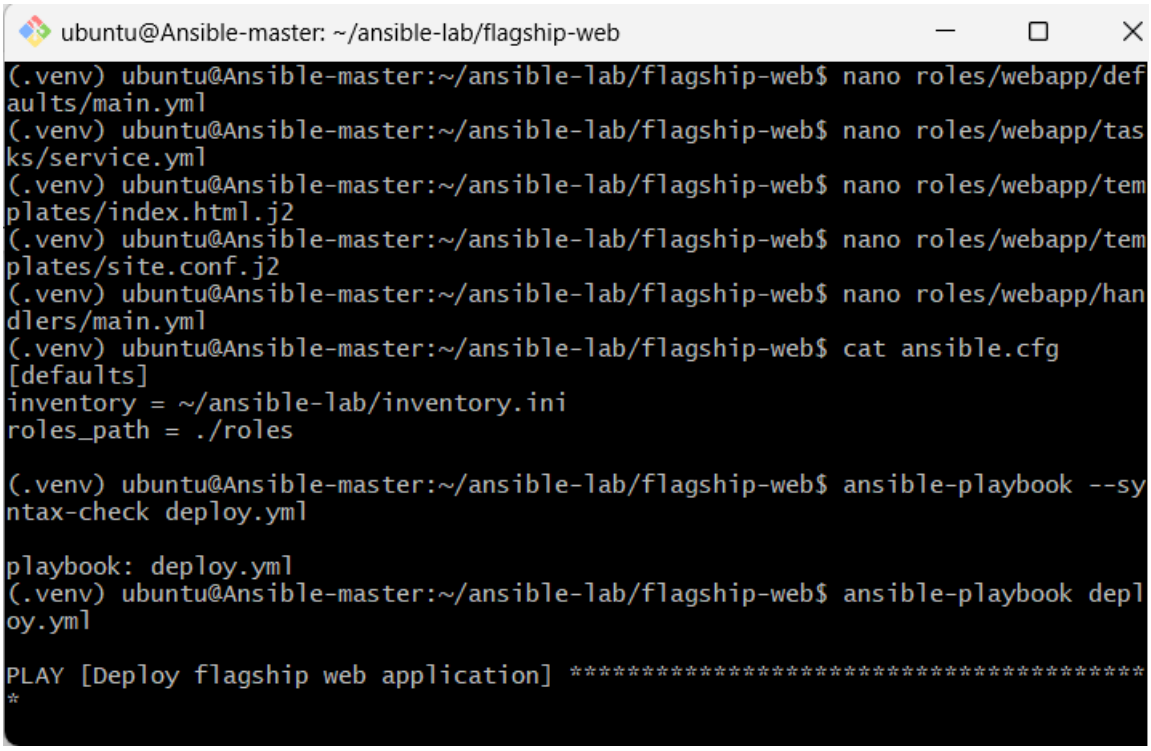

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
GNU nano 6.2 roles/webapp/tasks/service.yml
state: started
enabled: true
- name: Verify local http response
ansible.builtin.uri:
url: "http://127.0.0.1:{{ nginx_listen_port }}/"
status_code: 200
return_content: true
register: web_response
retries: 5
delay: 2
until: web_response.status == 200
- name: Display deployment verification
ansible.builtin.debug:
msg:
- "HTTP Status: {{ web_response.status }}"
- "Application name: {{ app_name }}"
- "Environment: {{ app_environment }}"
- "Version: {{ app_version }}"
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

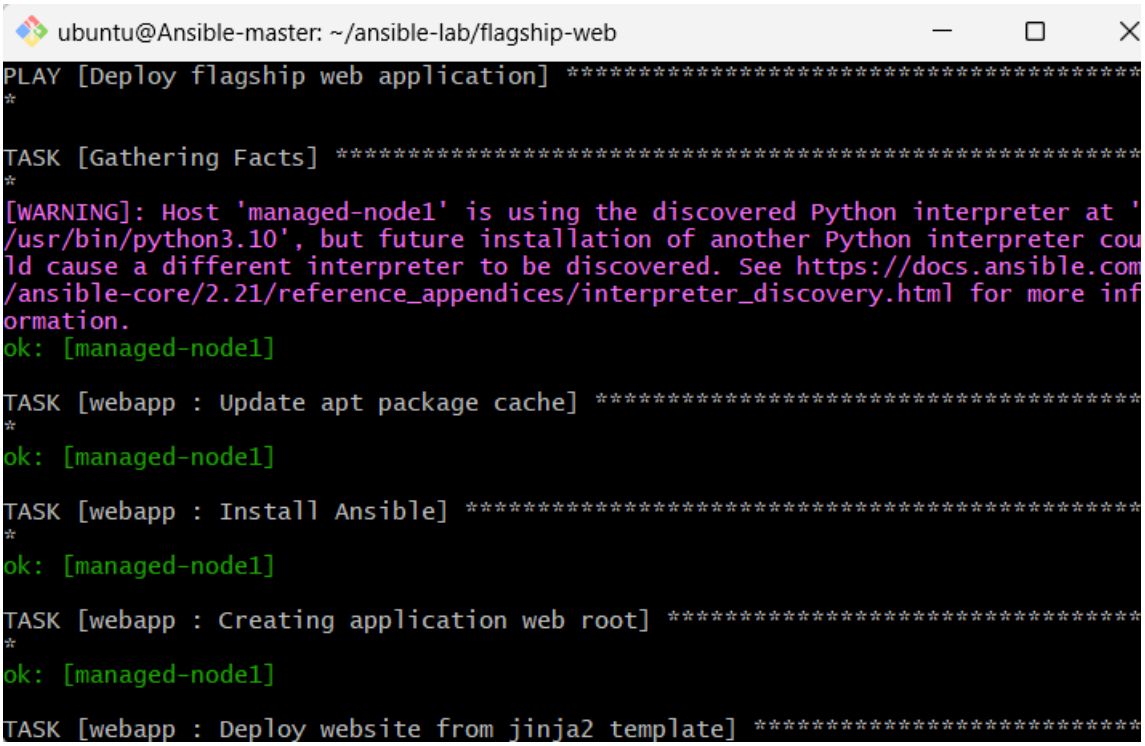

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
GNU nano 6.2 roles/webapp/templates/index.html.j2
<h1>{{ app_name }} is running!</h1>
Environment: {{ app_environment }}
Version: {{ app_version }}
Server: {{ ansible_facts['hostname'] }}
Operating System: {{ ansible_facts['distribution'] }}
<h2>Managed by Ansible and Vishnu</h2>
[Read 6 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

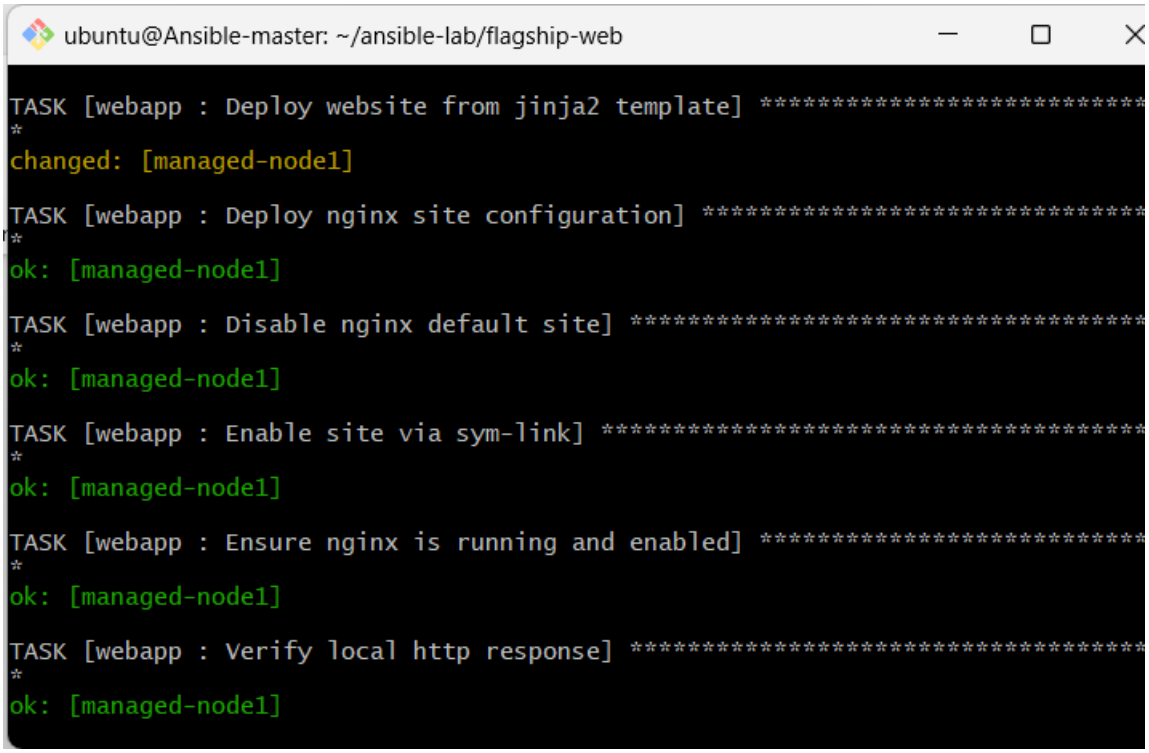

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
GNU nano 6.2 roles/webapp/templates/site.conf.j2
server {
listen {{ nginx_listen_port }} default_server;
listen [::]:{{ nginx_listen_port }} default_server;
server_name {{ nginx_server_name }};
root {{ web_root }};
index index.html;
location / {
try_files $uri $uri/ =404;
}
location = /health {
access_log off;
default_type text/plain;
return 200 "healthy\n";
}
location ~ /\. {
deny all;
}
}
[Read 18 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

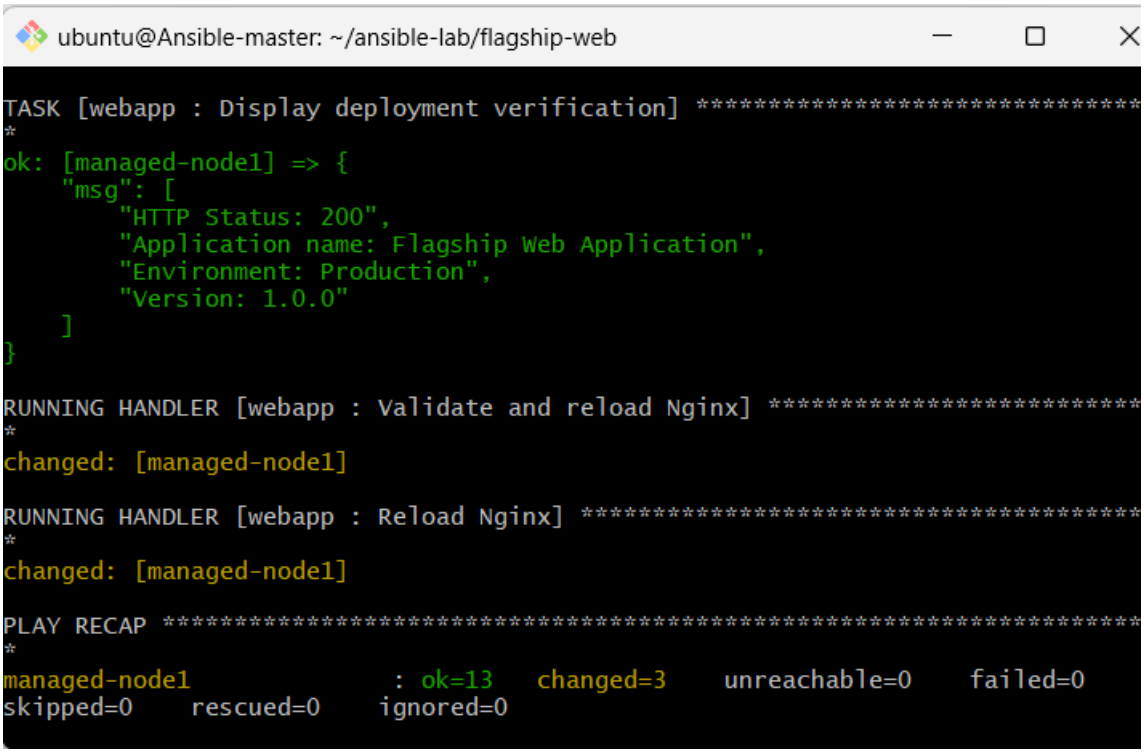

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
GNU nano 6.2 roles/webapp/handlers/main.yml

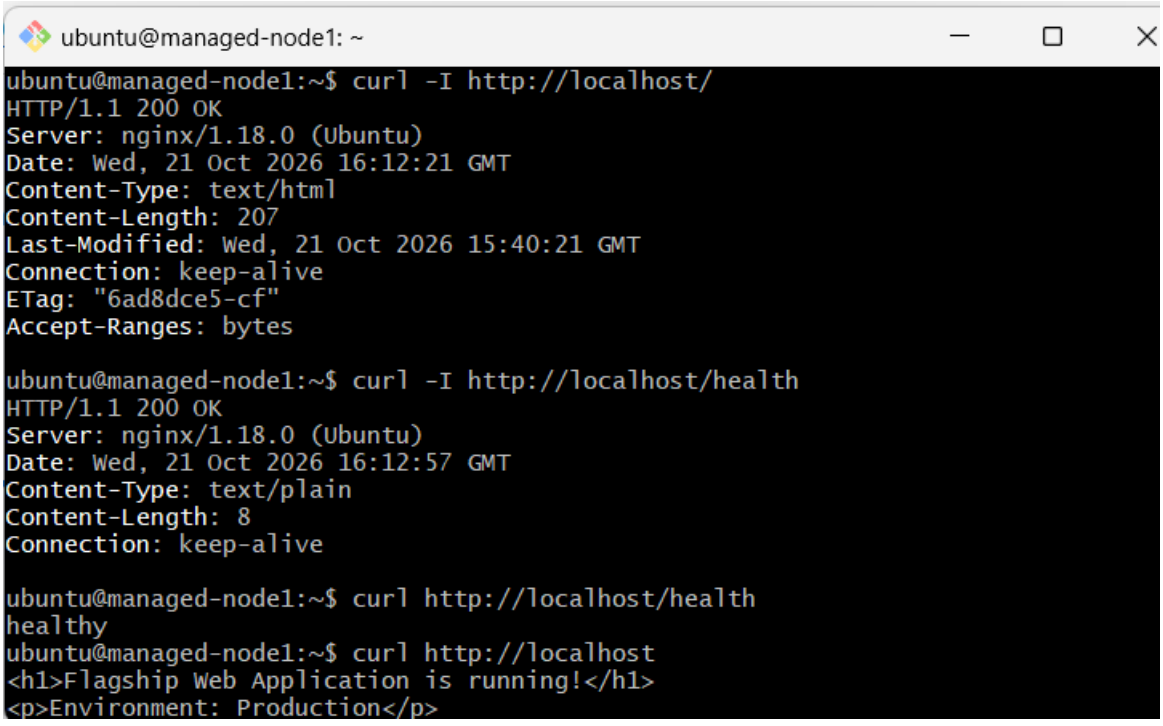
- name: Validate and reload Nginx
ansible.builtin.command:
argv:
- nginx
- -t
notify: Reload Nginx
- name: Reload Nginx
ansible.builtin.service:
name: nginx
state: reloaded
[Read 12 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

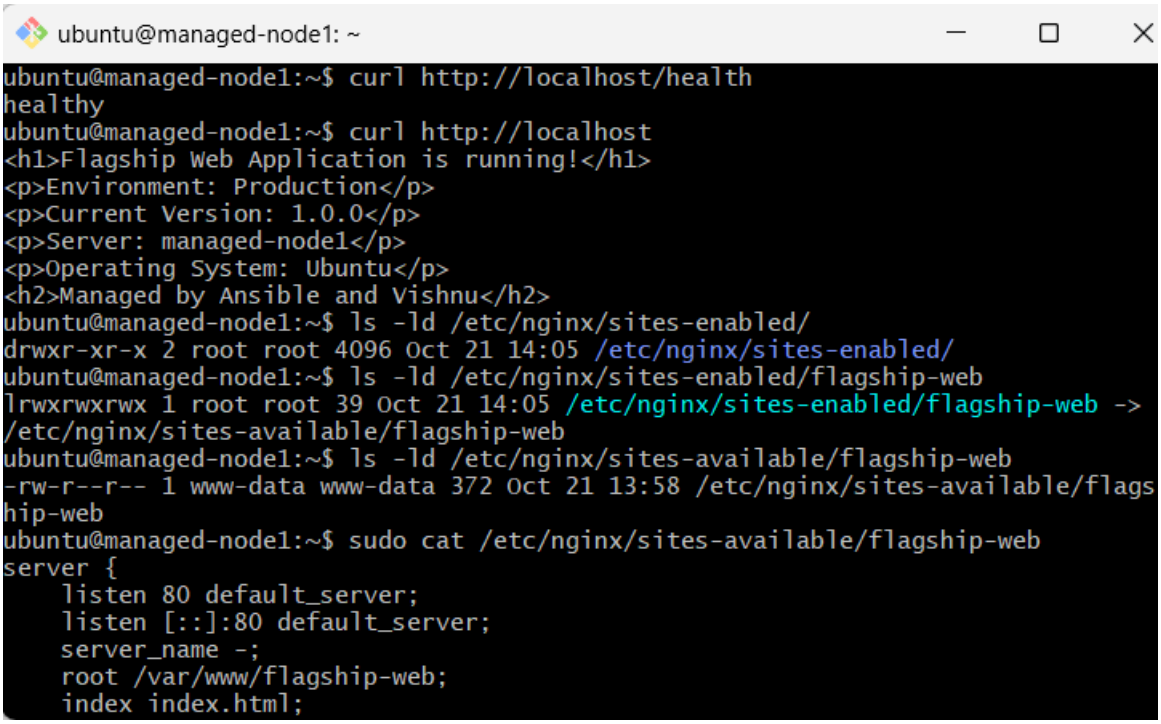

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-web$ nano roles/webapp/defaults/main.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-web$ nano roles/webapp/tasks/service.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-web$ nano roles/webapp/templates/index.html.j2
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-web$ nano roles/webapp/templates/site.conf.j2
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-web$ nano roles/webapp/handlers/main.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-web$ cat ansible.cfg
[defaults]
inventory = ~/ansible-lab/inventory.ini
roles_path = ./roles
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-web$ ansible-playbook --syntax-check deploy.yml
playbook: deploy.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-web$ ansible-playbook deploy.yml
PLAY [Deploy flagship web application] **
*


ubuntu@Ansible-master: ~/ansible-lab/flagship-web
PLAY [Deploy flagship web application] ***********************************
*
TASK [Gathering Facts] ***
*
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '
/usr/bin/python3.10', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com
/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more inf
ormation.
ok: [managed-node1]
TASK [webapp : Update apt package cache] *********************************
*
ok: [managed-node1]
TASK [webapp : Install Ansible] **
*
ok: [managed-node1]
TASK [webapp : Creating application web root] ****************************
*
ok: [managed-node1]
TASK [webapp : Deploy website from jinja2 template] **********************

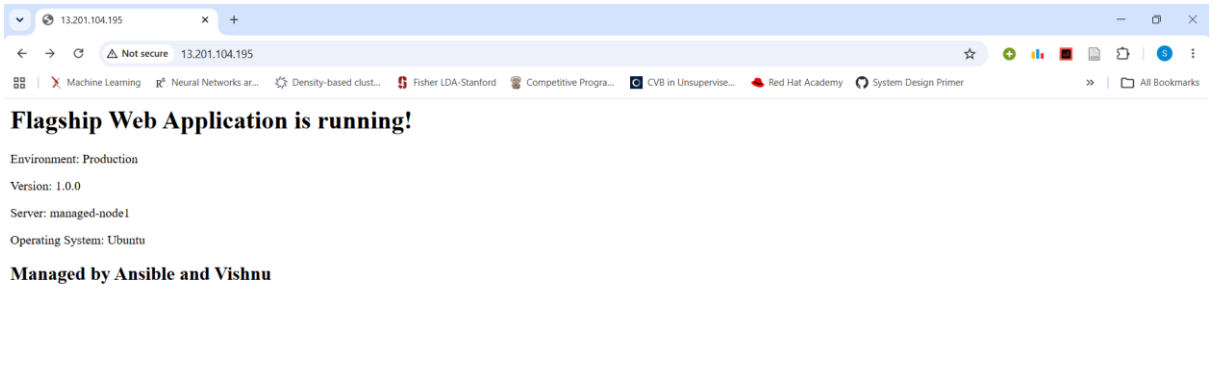

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
TASK [webapp : Deploy website from jinja2 template] **********************
*
changed: [managed-node1]
TASK [webapp : Deploy nginx site configuration] **************************
*
ok: [managed-node1]
TASK [webapp : Disable nginx default site] *******************************
*
ok: [managed-node1]
TASK [webapp : Enable site via sym-link] *********************************
*
ok: [managed-node1]
TASK [webapp : Ensure nginx is running and enabled] **********************
*
ok: [managed-node1]
TASK [webapp : Verify local http response] *******************************
*
ok: [managed-node1]

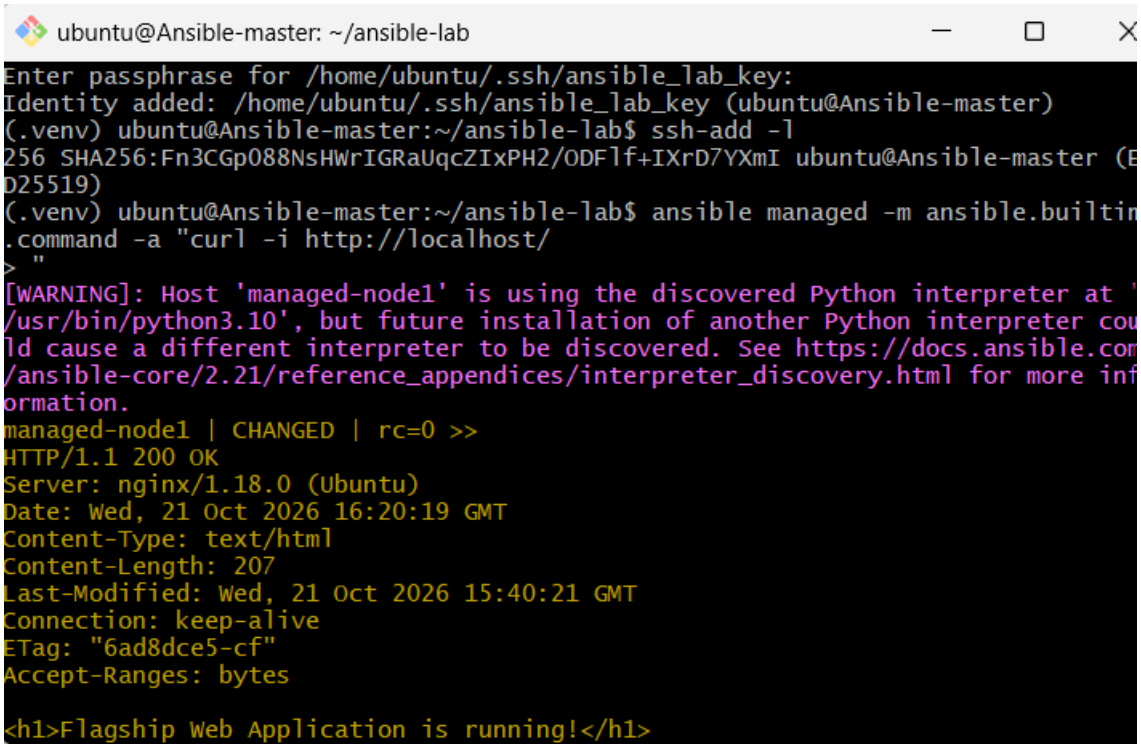

ubuntu@Ansible-master: ~/ansible-lab/flagship-web
TASK [webapp : Display deployment verification] ********************************
*
ok: [managed-node1] => {
"msg": [
"HTTP Status: 200",
"Application name: Flagship Web Application",
"Environment: Production",
"Version: 1.0.0"
]
}
RUNNING HANDLER [webapp : Validate and reload Nginx] ***************************
*
changed: [managed-node1]
RUNNING HANDLER [webapp : Reload Nginx] **
*
changed: [managed-node1]
PLAY RECAP ***
*
managed-node1 : ok=13 changed=3 unreachable=0 failed=0
skipped=0 rescued=0 ignored=0

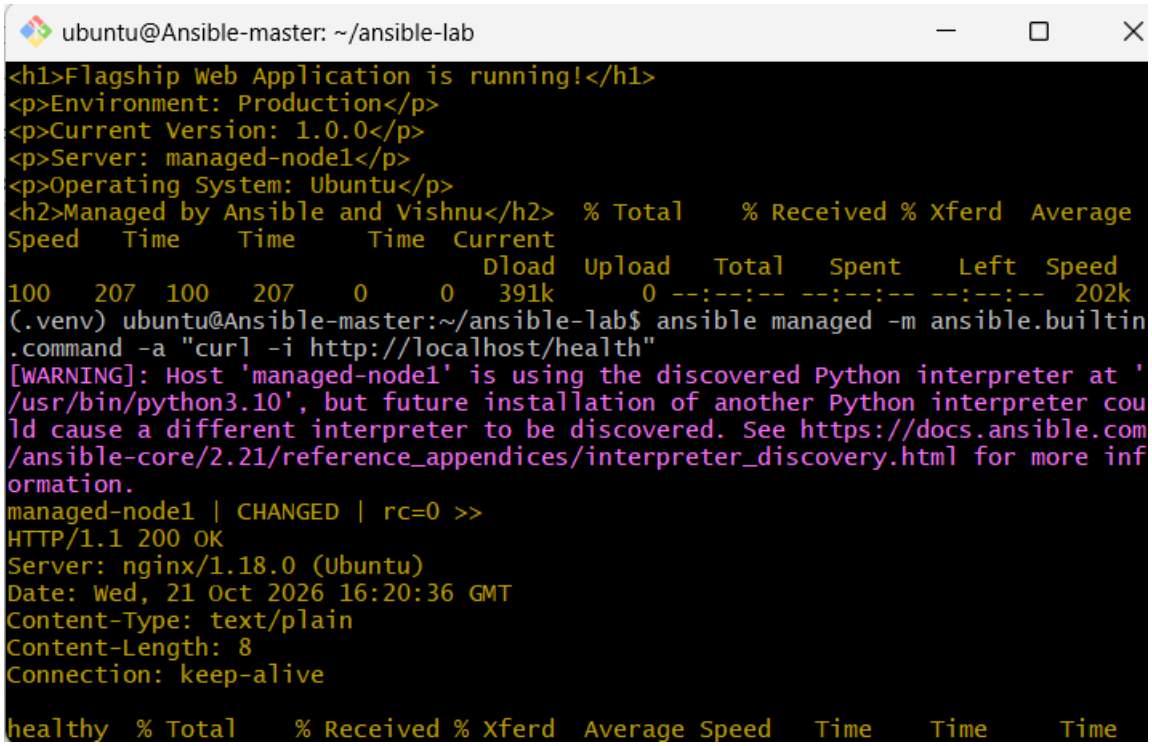

ubuntu@managed-node1: ~
ubuntu@managed-node1:~$ curl -I http://localhost/
HTTP/1.1 200 OK
Server: nginx/1.18.0 (Ubuntu)
Date: Wed, 21 Oct 2026 16:12:21 GMT
Content-Type: text/html
Content-Length: 207
Last-Modified: Wed, 21 Oct 2026 15:40:21 GMT
Connection: keep-alive
ETag: "6ad8dce5-cf"
Accept-Ranges: bytes
ubuntu@managed-node1:~$ curl -I http://localhost/health
HTTP/1.1 200 OK
Server: nginx/1.18.0 (Ubuntu)
Date: Wed, 21 Oct 2026 16:12:57 GMT
Content-Type: text/plain
Content-Length: 8
Connection: keep-alive
ubuntu@managed-node1:~$ curl http://localhost/health
healthy
ubuntu@managed-node1:~$ curl http://localhost
<h1>Flagship Web Application is running!</h1>
Environment: Production

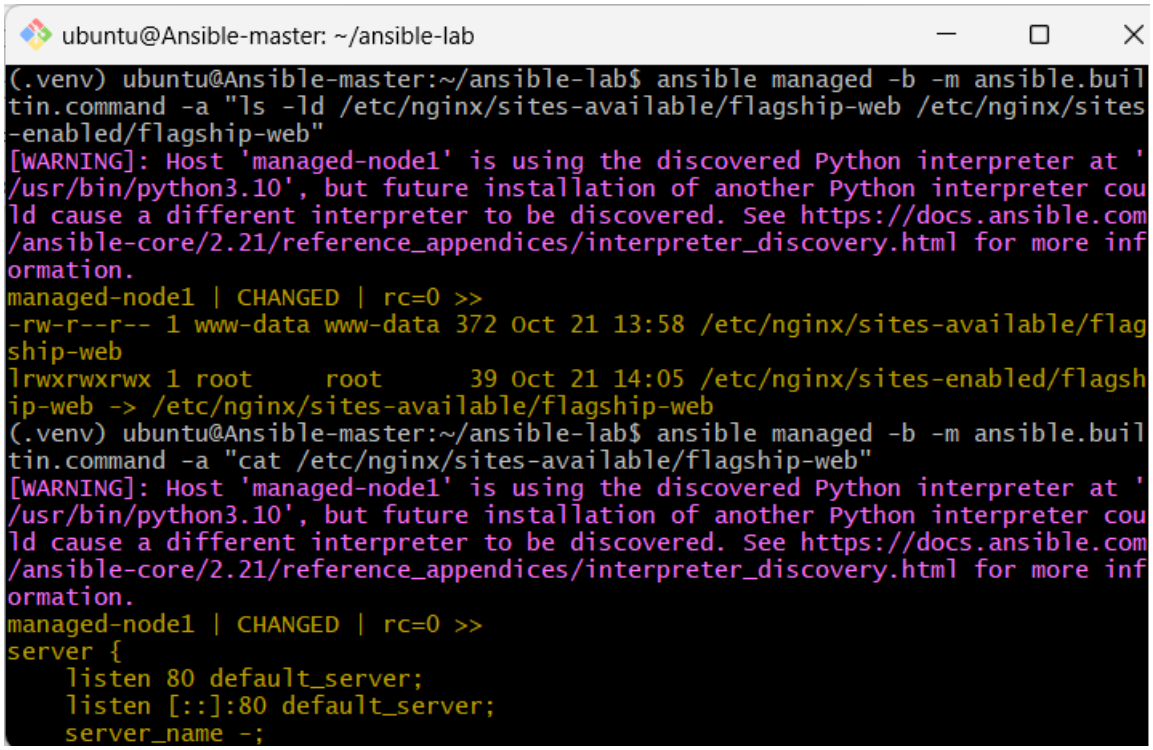

ubuntu@managed-node1: ~
ubuntu@managed-node1:~$ curl http://localhost/health
healthy
ubuntu@managed-node1:~$ curl http://localhost
<h1>Flagship Web Application is running!</h1>
Environment: Production
Current Version: 1.0.0
Server: managed-node1
Operating System: Ubuntu
<h2>Managed by Ansible and Vishnu</h2>
ubuntu@managed-node1:~$ ls -ld /etc/nginx/sites-enabled/
drwxr-xr-x 2 root root 4096 Oct 21 14:05 /etc/nginx/sites-enabled/
ubuntu@managed-node1:~$ ls -ld /etc/nginx/sites-enabled/flagship-web
lrwxrwxrwx 1 root root 39 Oct 21 14:05 /etc/nginx/sites-enabled/flagship-web ->
/etc/nginx/sites-available/flagship-web
ubuntu@managed-node1:~$ ls -ld /etc/nginx/sites-available/flagship-web
-rw-r--r-- 1 www-data www-data 372 Oct 21 13:58 /etc/nginx/sites-available/flags
hip-web
ubuntu@managed-node1:~$ sudo cat /etc/nginx/sites-available/flagship-web
server {
listen 80 default_server;
listen [::]:80 default_server;
server_name -;
root /var/www/flagship-web;
index index.html;

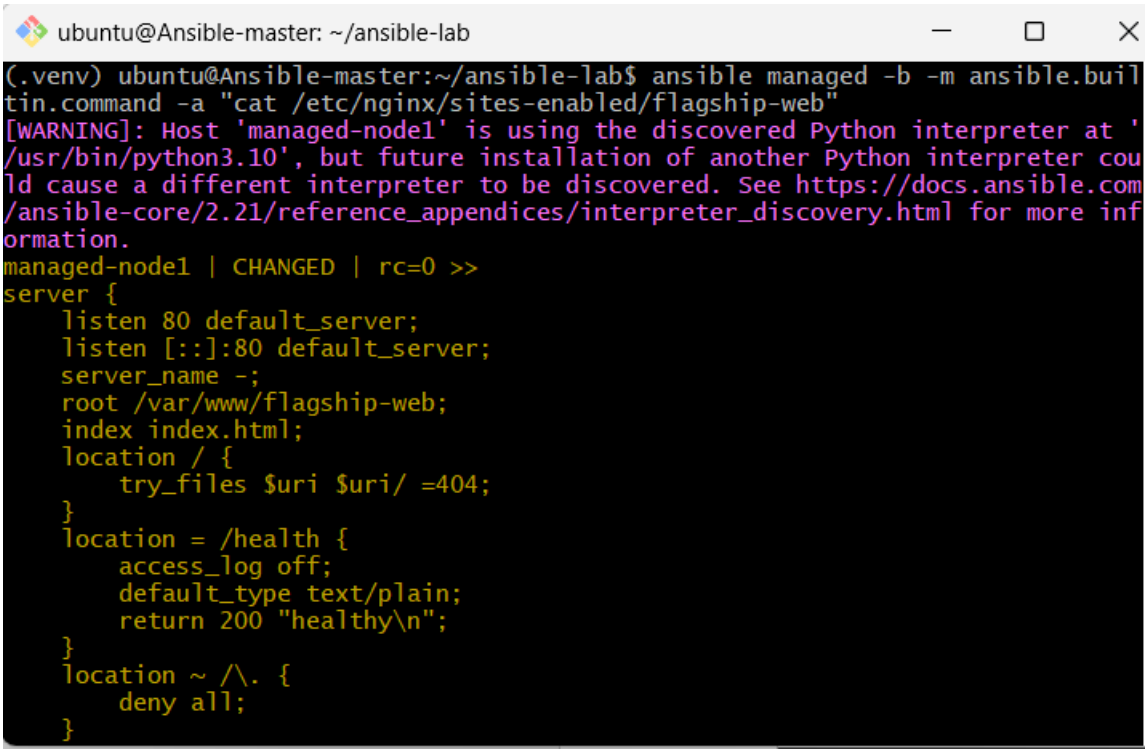

ubuntu@managed-node1: ~
ubuntu@managed-node1:~$ sudo cat /etc/nginx/sites-available/flagship-web
server {
listen 80 default_server;
listen [::]:80 default_server;
server_name -;
root /var/www/flagship-web;
index index.html;
location / {
try_files $uri $uri/ =404;
}
location = /health {
access_log off;
default_type text/plain;
return 200 "healthy\n";
}
location ~ /\. {
deny all;
}
}
ubuntu@managed-node1:~$ sudo cat /var/www/flagship-web/index.html
<h1>Flagship Web Application is running!</h1>
Environment: Production
Current Version: 1.0.0
Server: managed-node1

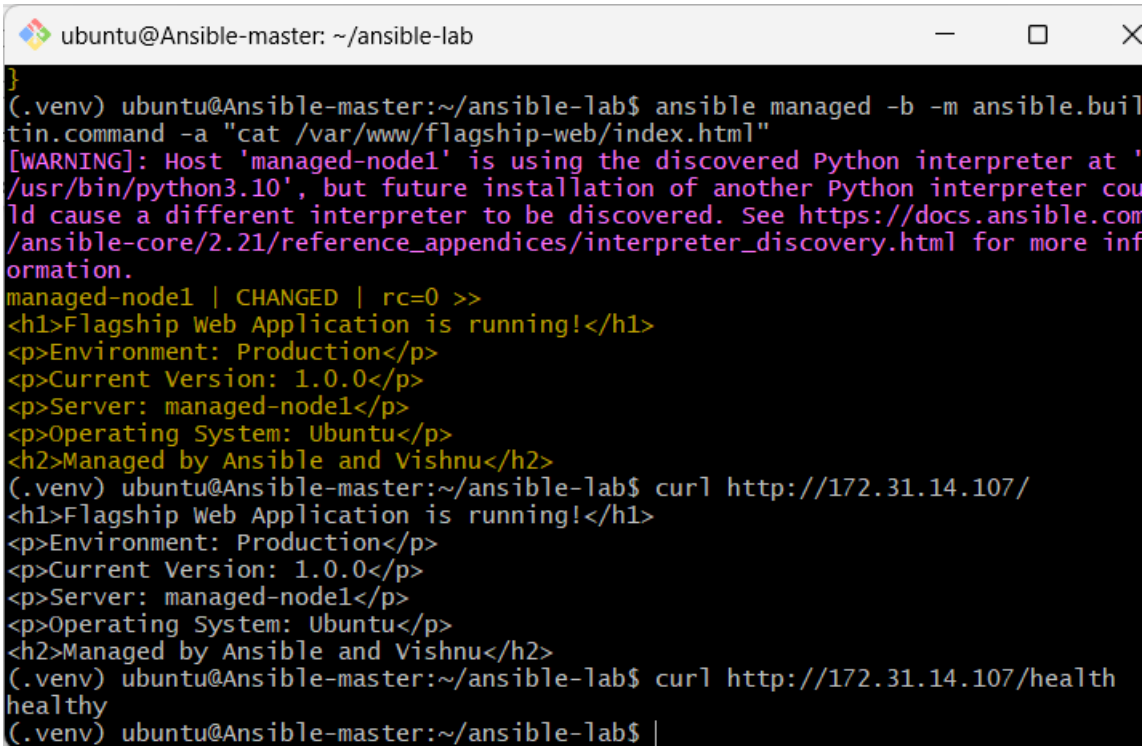

13.201.104.195
Not secure
13.201.104.195
Machine Learning
Neural Networks ar...
Density-based clust...
Fisher LDA-Stanford
Competitive Progra...
CVB in Unsupervise...
Red Hat Academy
System Design Primer
All Bookmarks
Flagship Web Application is running!
Environment: Production
Version: 1.0.0
Server: managed-node1
Operating System: Ubuntu
Managed by Ansible and Vishnu

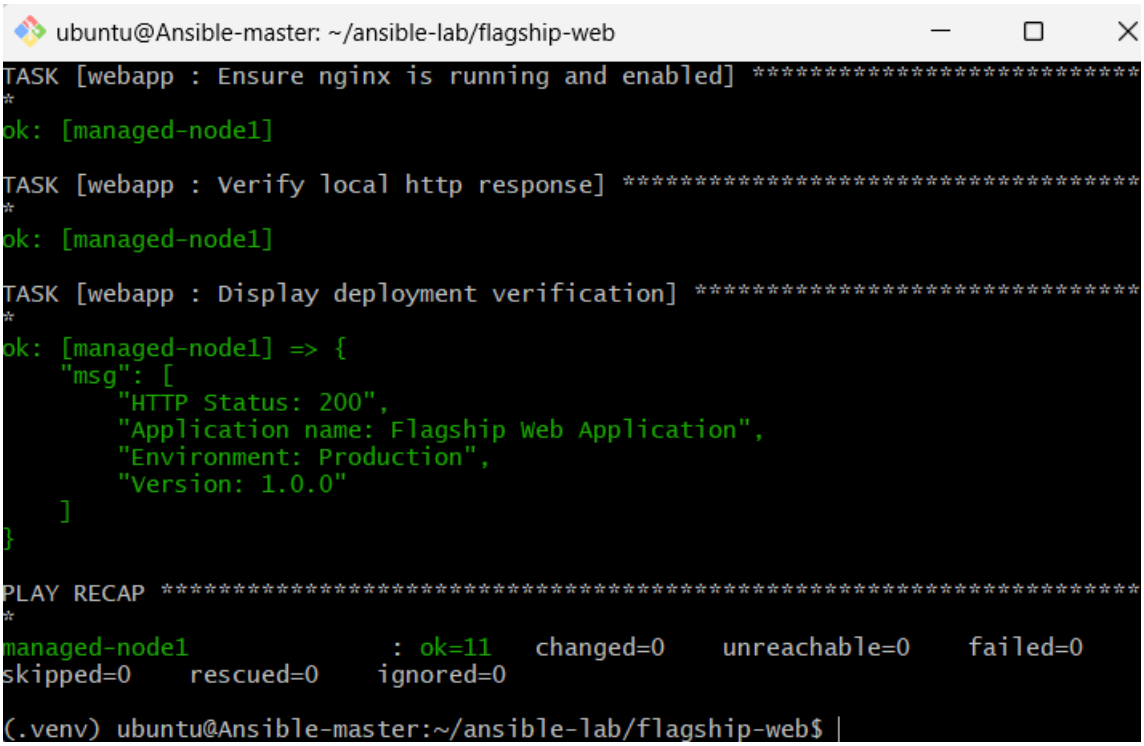

ubuntu@Ansible-master: ~/ansible-lab
Enter passphrase for /home/ubuntu/.ssh/ansible_lab_key:
Identity added: /home/ubuntu/.ssh/ansible_lab_key (ubuntu@Ansible-master)
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ssh-add -l
256 SHA256:Fn3CGpO88NsHWrIGRaUqcZIxPH2/ODFlf+IXrD7YXmI ubuntu@Ansible-master (E
D25519)
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -m ansible.builtin
.command -a "curl -i http://localhost/
> "
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '
/usr/bin/python3.10', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com
/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more inf
ormation.
managed-node1 | CHANGED | rc=0 >>
HTTP/1.1 200 OK
Server: nginx/1.18.0 (Ubuntu)
Date: Wed, 21 Oct 2026 16:20:19 GMT
Content-Type: text/html
Content-Length: 207
Last-Modified: Wed, 21 Oct 2026 15:40:21 GMT
Connection: keep-alive
ETag: "6ad8dce5-cf"
Accept-Ranges: bytes
<h1>Flagship Web Application is running!</h1>


ubuntu@Ansible-master: ~/ansible-lab
<h1>Flagship Web Application is running!</h1>
Environment: Production
Current Version: 1.0.0
Server: managed-node1
Operating System: Ubuntu
<h2>Managed by Ansible and Vishnu</h2> % Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 207 100 207 0 0 391k 0 --:--:-- --:--:-- --:--:-- 202k
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -m ansible.builtin.command -a "curl -i http://localhost/health"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
managed-node1 | CHANGED | rc=0 >>
HTTP/1.1 200 OK
Server: nginx/1.18.0 (Ubuntu)
Date: Wed, 21 Oct 2026 16:20:36 GMT
Content-Type: text/plain
Content-Length: 8
Connection: keep-alive
healthy % Total % Received % Xferd Average Speed Time Time Time


ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -b -m ansible.builtin.command -a "ls -ld /etc/nginx/sites-available/flagship-web /etc/nginx/sites-enabled/flagship-web"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
managed-node1 | CHANGED | rc=0 >>
-rw-r--r-- 1 www-data www-data 372 Oct 21 13:58 /etc/nginx/sites-available/flagship-web
lrwxrwxrwx 1 root root 39 Oct 21 14:05 /etc/nginx/sites-enabled/flagship-web -> /etc/nginx/sites-available/flagship-web
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -b -m ansible.builtin.command -a "cat /etc/nginx/sites-available/flagship-web"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
managed-node1 | CHANGED | rc=0 >>
server {
listen 80 default_server;
listen [::]:80 default_server;
server_name _;


ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -b -m ansible.buil
tin.command -a "cat /etc/nginx/sites-enabled/flagship-web"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '
/usr/bin/python3.10', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com
/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more inf
ormation.
managed-node1 | CHANGED | rc=0 >>
server {
listen 80 default_server;
listen [::]:80 default_server;
server_name -;
root /var/www/flagship-web;
index index.html;
location / {
try_files $uri $uri/ =404;
}
location = /health {
access_log off;
default_type text/plain;
return 200 "healthy\n";
}
location ~ /\. {
deny all;
}


ubuntu@Ansible-master: ~/ansible-lab
}
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -b -m ansible.buil
tin.command -a "cat /var/www/flagship-web/index.html"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '
/usr/bin/python3.10', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com
/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more inf
ormation.
managed-node1 | CHANGED | rc=0 >>
<h1>Flagship Web Application is running!</h1>
Environment: Production
Current Version: 1.0.0
Server: managed-node1
Operating System: Ubuntu
<h2>Managed by Ansible and Vishnu</h2>
(.venv) ubuntu@Ansible-master:~/ansible-lab$ curl http://172.31.14.107/
<h1>Flagship Web Application is running!</h1>
Environment: Production
Current Version: 1.0.0
Server: managed-node1
Operating System: Ubuntu
<h2>Managed by Ansible and Vishnu</h2>
(.venv) ubuntu@Ansible-master:~/ansible-lab$ curl http://172.31.14.107/health
healthy
(.venv) ubuntu@Ansible-master:~/ansible-lab$



Finally, last but not the least “idempotency” as we go